30. CaO and NaCl have the same crystal structure and approximately the same radii. It U is the lattice energy of NaCl, the approximate lattice energy of CaO is

a) U b) 2U

c) U/2 d) 4U

II. Write brief answer to the following questions.

31. Define the following

i) Bond order

ii) Hybridisation

iii) σ- bond

32. What is a pi bond?

33. In CH_4, NH_3 and H_2O, the central atom undergoes sp^3 hybridisation - yet their bond angles are different. why?

34. Explain Sp^2 hybridisation in BF_3

35. Draw the M.O diagram for oxygen molecule calculate its bond order and show that O_2 is paramagnetic.

36. Draw MO diagram of CO and calculate its bond order.

37. What do you understand by Linear combination of atomic orbitals in MO theory.

38. Discuss the formation of N_2 molecule using MO Theory

39. What is dipolemoment?

40. Linear form of carbondioxide molecule has two polar bonds. yet the molecule has Zero dipolement why?

41. Draw the Lewis structures for the following species.

i) NO_3^- ii) SO_4^{2-}

iii) HNO_3 iv) O_3

42. Explain the bond formation in $BeCl_2$ and $MgCl_2$.

43. Which bond is stronger σ or π? Why?

44. Define bond energy.

45. Hydrogen gas is diatomic where as inert gases are monoatomic – explain on the basis of MO theory.

46. What is Polar Covalent bond? explain with example.

47. Considering x- axis as molecular axis, which out of the following will form a sigma bond.

i) 1s and $2p_y$ ii) $2P_x$ and $2P_x$

iii) $2p_x$ and $2p_z$ iv) 1s and $2p_z$

48. Explain resonance with reference to carbonate ion?

49. Explain the bond formation in ethylene and acetylene.

50. What type of hybridisations are possible in the following geometeries?

a) octahedral

b) tetrahedral

c) square planer.

51. Explain VSEPR theory. Applying this theory to predict the shapes of IF_7, and SF_6

52. CO_2 and H_2O both are triatomic molecule but their dipole moment values are different. Why?

53. Which one of the following has highest bond order?

N_2, N_2^+ or N_2^-

54. Explain the covalent character in ionic bond.

55. Describe fajan's rule.

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